

Machine Learning Project Presentation Guidelines

Course: DS 612: Data Mining with Business Applications

Time Limit: Aim for 5–10 minutes per group; the more concise the better

SUGGESTED OUTLINE

NOTE: You don't have to follow the outline below. However, all groups should leave time for Q&A. Part of the presentation grade will be based on Q&A. Focus on those that are likely to be appealing to non-technical audiences. For example, any interesting business insights your team have gleaned at the end of your analysis?

1. Title Slide

- Project title
 - Names of all group members
 - Date
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2. Problem Statement

- Describe the business or real-world problem you are addressing
 - Why is this problem important?
 - Is the goal to classify (categorical output) or regress (quantitative output)?
 - Clearly define your prediction target variable
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3. Data Overview

- Source of data (e.g., Kaggle, UCI)
 - Number of features and instances (ensure they are within the 10–100 features, 1,000–10,000 instances)
 - Summary statistics: distributions, missing data, outliers
 - Visualizations: Histograms, correlation matrix, scatter plots (if helpful)
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4. Data Preparation

- How did you clean the data? (handling missing values, outliers)
- Feature engineering or transformation

- Did you normalize or scale your features? Why or why not?
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5. Model Selection

- Which machine learning models did you try (e.g., Ridge Regression, Logistic Regression, Decision Trees, SVM, Random Forest, etc.)?
 - Why did you choose these models?
 - Include basic explanation (business audience appropriate)
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6. Model Training & Evaluation

- How did you split your data (training vs testing)?
 - Any cross-validation used?
 - Show performance metrics (accuracy, Rsquare, error rate, etc., depending on problem type)
 - Include comparison charts or tables
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7. Parameter Tuning

- Briefly explain any hyperparameter tuning you did
 - Methods used (e.g., Grid Search, manual tuning)
 - Show how performance improved (if applicable)
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8. Underfitting / Overfitting

- Did your models suffer from underfitting or overfitting?
 - How did you detect it (e.g., train/test performance discrepancy)?
 - What steps did you take to address it?
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9. Results & Insights

- Which model performed best and why?
 - Key findings or patterns discovered in the data
 - Any surprising or disappointing results? What might explain them?
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10. Code Demonstration (brief)

- Walk through a key part of your Jupyter Notebook (preprocessing, model training, evaluation)
 - Code should run cleanly with comments and explanations
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11. Lessons Learned & Recommendations

- What did you learn about data mining and machine learning?
 - Reflect on what worked well vs. what didn't
 - Would you recommend a specific algorithm for this kind of task based on your experience?
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12. Q&A (if live) / Closing Remarks (if recorded)

- Encourage questions or briefly summarize your key takeaway